ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ 1 SET B **WINTER SEMESTER, 2024-2025**

DURATION: 30 MINUTES **FULL MARKS: 20**

MATH 4341: Linear Algebra

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. The *UpSiDe DoWn* is a *strange* world where every vector from the real world appears shifted, rotated or rearranged in unexpected ways. Mr.Harrington and Mr.Henderson, two curious explorers studying this strange realm have discovered pairs of vectors that correspond between the two worlds. They observed $\mathbf{v_1} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $\mathbf{v_2} = \begin{bmatrix} -2 \\ -1 \end{bmatrix}$ in the real world correspond to $\mathbf{u_1} = \begin{bmatrix} -1 \\ -5 \end{bmatrix}$ and $\mathbf{u_2} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$ in the *UpSiDe DoWn* world respectively. And it is obvious that the standard basis vectors in the real world are $\hat{\mathbf{i}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$(for x-axis) and $\hat{\mathbf{j}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$(for y-axis). 3 + 2 (CO2) (PO1)

   (a) Determine the basis vectors of the *UpSiDe DoWn* world (i.e., the transformation matrix that converts real-world vectors into *UpSiDe DoWn* vectors).

   (b) Using your result from part (a), answer the following:

   (b)i. Objects in the *UpSiDe DoWn* world are: ☐ same ☐ enlarged ☐ squeezed

   (b)ii. The orientation in the *UpSiDe DoWn* world is: ☐ as usual ☐ flipped

NB: Your answers for parts (b)(i) and (b)(ii) should be supported by brief mathematical justification. It is not necessary to show the steps of solving equations for this question.

Solution:

(a) Let T be the transformation matrix such that $T\mathbf{v}_1 = \mathbf{u}_1$ and $T\mathbf{v}_2 = \mathbf{u}_2$. We are given that:

$$T\begin{bmatrix} 3 & -2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ -5 & 2 \end{bmatrix}.$$

Let $T = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Therefore, we need to solve:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 3 & -2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ -5 & 2 \end{bmatrix}.$$

Expanding the left-hand side:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 3 & -2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 3a+b & -2a-b \\ 3c+d & -2c-d \end{bmatrix}.$$

Equating the result with the right-hand side:

$$\begin{bmatrix} 3a+b & -2a-b \\ 3c+d & -2c-d \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ -5 & 2 \end{bmatrix}.$$

This gives the following system of equations:

(i) $3a + b = -1$

(ii) $-2a - b = 0$

(iii) $3c + d = -5$

(iv) $-2c - d = 2$

Using a calculator to solve this system, we find:

$$a = -1, \quad b = 2, \quad c = -3, \quad d = 4.$$

Therefore, the transformation matrix T is:

$$T = \begin{bmatrix} -1 & 2 \\ -3 & 4 \end{bmatrix}.$$

The basis vectors are the columns: $\begin{bmatrix} -1 \\ -3 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$.

(b) (i) $\det(A) = (-1)(4) - (2)(-3) = -4 + 6 = 2$. Since $|\det(A)| > 1$, the objects are **enlarged**.
(ii) Since $\det(A) > 0$, the orientation is **as usual**.

(NB): You can also find the inverse of the transformation matrix to determine the basis vectors of the real world in terms of the *UpSiDe DoWn* world.

Final Answers:

$$1(a)\ T = \begin{bmatrix} -1 & 2 \\ -3 & 4 \end{bmatrix}$$

1(b) (i) Enlarged, (ii) Usual

Rubrics:

- 1 mark for setting up the Transformation equations correctly.
- 1 mark for solving the equations correctly to find the transformation matrix.
- 1 mark for correctly determining the bases of the transformation matrix.
- 1 mark for correctly calculating the determinant of T.
- 1 mark for correctly commenting on the change in shape and orientation.

2. You are provided with the following equations - 1 + 0.5 + 5 + 1 (CO2) (PO1)

$$x + 2y - z = 5 \tag{1}$$
$$2x + 4y + 2z = 6 \tag{2}$$
$$3x - y + 2z = 3 \tag{3}$$

(a) Write down the system of linear equations in $Ax = b$ or Matrix form.

(b) Write down the augmented matrix ($[A|b]$).

(c) Perform necessary operations on the coefficient matrix $\mathbf{A}$ to obtain the upper triangular matrix $\mathbf{U}$ using elimination matrices $\mathbf{E}_{ij}$. Convert $\mathbf{A}x = \mathbf{b}$ to $\mathbf{U}x = \mathbf{c}$ (where $\mathbf{U}$ is an upper triangular matrix).

(d) Find out the values of x,y and z using back substitution.

Solution:

We begin by swapping row 2 and row 3 to prevent potential issues during elimination. This can be done by using the permutation matrix P_{23}.

$$x + 2y - z = 5 \tag{4}$$
$$2x + 4y + 2z = 6 \tag{5}$$
$$3x - y + 2z = 3 \tag{6}$$

a) We can now represent the system of equations in matrix form $Ax = b$:

$$\underset{\text{A}}{\begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & 2 \\ 3 & -1 & 2 \end{bmatrix}} \underset{\text{x}}{\begin{bmatrix} x \\ y \\ z \end{bmatrix}} = \underset{\text{b}}{\begin{bmatrix} 5 \\ 6 \\ 3 \end{bmatrix}}$$

b) The augmented matrix $[A|b]$ for the system is:

$$\underset{\text{A}}{\begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & 2 \\ 3 & -1 & 2 \end{bmatrix}} \underset{|}{\Big|} \underset{\text{b}}{\begin{bmatrix} 5 \\ 6 \\ 3 \end{bmatrix}}$$

c) Now, let's swap row 2 and row 3. This can be achieved by applying the permutation matrix P_{23}, which swaps rows 2 and 3:

$$P_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Applying P_{23} to the augmented matrix:

$$P_{23} \cdot \left[\begin{array}{ccc|c} 1 & 2 & -1 & 5 \\ 2 & 4 & 2 & 6 \\ 3 & -1 & 2 & 3 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 2 & -1 & 5 \\ 3 & -1 & 2 & 3 \\ 2 & 4 & 2 & 6 \end{array}\right]$$

Now, apply the elimination matrix E_{21} to eliminate the element in row 2, column 1:

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Applying E_{21} to the augmented matrix:

$$E_{21} \cdot \left[\begin{array}{ccc|c} 1 & 2 & -1 & 5 \\ 3 & -1 & 2 & 3 \\ 2 & 4 & 2 & 6 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 2 & -1 & 5 \\ 0 & -7 & 5 & -12 \\ 2 & 4 & 2 & 6 \end{array}\right]$$

Next, apply the elimination matrix E_{31} to eliminate the element in row 3, column 1:

$$E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

Applying E_{31} to the augmented matrix:

$$E_{31} \cdot \left[\begin{array}{ccc|c} 1 & 2 & -1 & 5 \\ 0 & -7 & 5 & -12 \\ 4 & -6 & 4 & -2 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 2 & -1 & 5 \\ 0 & -7 & 5 & -12 \\ 0 & 0 & 4 & -4 \end{array}\right]$$

Now we have the upper triangular matrix:

$$\begin{array}{c} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 5 \\ 0 & -7 & 5 & -12 \\ 0 & 0 & 4 & -4 \end{array}\right] \\ \text{U} \qquad\qquad \text{x} \qquad \text{c} \end{array}$$

d) To solve for x, y, and z, we now perform back substitution:

From the third equation:

$$4z = -4 \Rightarrow z = \frac{-4}{4} = -1$$

Substituting $z = -1$ into the second equation:

$$-7y + 5z = 12 \Rightarrow -7y + 5(-1) = -12$$

Solving for y:

$$-7y + 5(-1) = -12 \Rightarrow -7y - 5 = -12 \Rightarrow -7y = -12 + 5 = -7 \Rightarrow y = \frac{-7}{-7} = 1$$

Now, substitute $z = -1$ and $y = 1$ into the first equation:

$$x + 2(1) - 1(-1) = 5$$

Solving for x:

$$x + 2(1) - 1(-1) = 5 \Rightarrow x + 2 + 1 = 5 \Rightarrow x + 3 = 5 \Rightarrow x = 5 - 3 = 2$$

$$x = 2$$

Final Answers:

x = 2, y = 1, z = -1

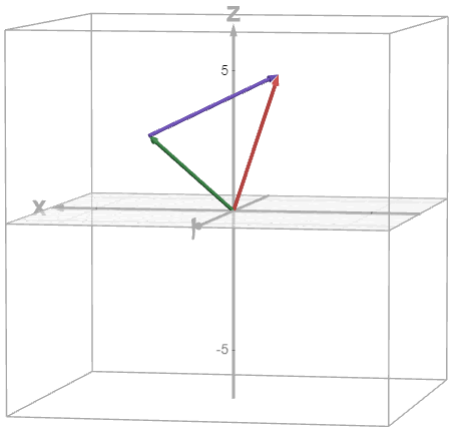
Rubrics:

- 1 mark for writing down the system of linear equations in $Ax = b$ or Matrix form.
- 0.5 mark for writing down the augmented matrix.

- 0.5 mark for dealing with temporary failure.
- 1.5 marks for performing the correct elimination steps to form an upper triangular matrix.
- 1 mark for correctly determining the values of x, y, and z.

3. Answer the following question by ticking the correct box (Question 3.a - Question 3.d) and writing the correct answer in the box (Question 3.c). 5 (CO2) (PO1)

a) What type of picture is shown below?



☐ Row Picture ☐ Column Picture

b) What is the span of two vectors $\vec{a}$ and $-2\vec{a}$?
☐ 3D space ☐ 2D space ☐ Line ☐ $(0,0)$

$$H = \begin{bmatrix} 161/23 & 21 & -28 \\ 527/17 & 93 & -124 \\ 119/17 & 21 & -28 \end{bmatrix}$$

c) What is the span of the columns of the matrix H?
☐ 3D space ☐ 2D space
☐ A line ☐ Origin

d) Which are the independent columns of the matrix H? (Multiple ticks are allowed)
☐ 1st ☐ 2nd ☐ 3rd ☐ None

e) What is the trace of the matrix H?
Answer:

Solution:

Final Answers:

(a) The picture shows a **Column Picture**. (Vectors have been shown in 3D space.)

(b) The span of the vectors $\vec{a}$ and $-2\vec{a}$ is a **Line** passing through the origin. (Both vectors are scalar multiples of each other.)

(c) The span of the columns of the matrix H is a **Line**. (The rank of the matrix is 1)

(d) The independent columns of the matrix H is the **1st column only**. (The rank of the matrix is 1)

(e) The trace of the matrix H is $(\frac{161}{23} + 93 + -28) = 72$.

Rubrics:

- 1 mark for each correct answer.

4. Based on the following row operations **(applied in order)** to a matrix B of order 3×3, determine 2.5 (CO2) (PO1)

(a) The **elimination matrix** E that transforms B into the upper triangular matrix U.

(b) Its inverse, E^{-1} (i.e., the corresponding lower triangular matrix L).

(i) $r_2 = r_2 - 3 \times r_1$

(ii) $r_3 = r_3 - 1 \times r_1$

(iii) $r_3 = r_3 - 2 \times r_2$

Solution:

Let the elementary elimination matrices corresponding to the given row operations be:

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}.$$

The elimination matrix E that transforms B into U is the product of these in reverse order of application:

$$E = E_3 E_2 E_1.$$

Carry out the multiplication step by step:
First, $E_2 E_1$:

$$E_2 E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}.$$

Now, $E = E_3(E_2 E_1)$:

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 5 & -2 & 1 \end{bmatrix}.$$

Thus:

(a) The elimination matrix,

$$E = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 5 & -2 & 1 \end{bmatrix}.$$

(b) The inverse of elimination matrix is (taking negative of the off-diagonal elements):

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & 2 & 1 \end{bmatrix}.$$

Final Answers:

(a)

$$E = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 7 & -2 & 1 \end{bmatrix}.$$

(b)

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & 2 & 1 \end{bmatrix}.$$

Rubrics:

- 2 mark for each correctly determining E.
- 0.5 mark for each correctly determining L.